

LitSense Progress Report

Date: December 8, 2025

Agents in the System

1. Interaction Analysis Agent
 2. Discussion Analysis Agent
 3. Role Assignment Agent
 4. Matching Agent
 5. Progress Tracking Agent
 6. AI Tutor Agent
 7. Smart Notifications Agent
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Current Agents Implemented

1. Interaction Analysis Agent

Description: Monitors how users interact with the PDF by tracking highlights, time spent on sections, and eye tracking data to detect when someone is struggling with content.

Goal: Automatically identify when a researcher is having difficulty understanding a paper section so the system can offer appropriate help.

Features:

- Runs monitoring checks every 5 seconds
- Calculates understanding scores from 0-100 based on interaction patterns
- Tracks highlight frequency, eye gaze patterns, and time spent per section
- Emits "struggle-detected" events when understanding score drops below 40
- Records all interaction data for later analysis

Communication with other agents:

- Sends struggle events to Discussion Analysis Agent
- Provides user understanding scores to Matching Agent
- Sends interaction logs to Progress Tracking Agent
- Shares section focus data with Role Assignment Agent

Prompts:

- "Has the user spent more than 3 minutes on this section with minimal progress?"
- "Are confused highlights increasing in frequency within the same section?"
- "Is the eye tracking showing repeated back-and-forth reading patterns indicating confusion?"
- "Based on time spent, highlight patterns, and eye tracking: calculate understanding score and determine if intervention is needed"

2. Discussion Analysis Agent

Description: Facilitates peer-to-peer help by managing chat conversations between struggling students and helpers.

Goal: Enable effective knowledge transfer by creating structured conversations between people who need help and those who can provide it.

Features:

- Opens real-time chat channels via Socket.io
- Generates conversation starters based on the confused content
- Manages discussion flow and keeps conversations focused
- Tracks conversation quality and outcomes
- Supports multiple simultaneous help sessions

Communication with other agents:

- Receives struggle notifications from Interaction Analysis Agent
- Gets peer match recommendations from Matching Agent
- Sends discussion outcomes to Progress Tracking Agent
- Works with Smart Notifications Agent to invite participants

Prompts:

- "Given the confused section and helper's expertise, what's the most effective conversation starter?"
 - "Is the discussion addressing the core confusion or drifting off-topic?"
 - "Has the struggler shown understanding improvement after 5 minutes of discussion?"
 - "Should we escalate to AI help if peer discussion isn't resolving the confusion?"
 - "What follow-up questions would help the helper understand the struggler's knowledge gap?"
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3. Role Assignment Agent

Description: Assigns specific PDF sections to team members so everyone knows what part they're responsible for reading and understanding.

Goal: Organize collaborative reading by distributing sections among team members and tracking who owns what.

Features:

- Allows manual section assignment by team leaders
- Tracks which user is assigned to which sections
- Broadcasts assignments in real-time to all team members
- Shows section ownership in the interface
- Supports reassignment when needed

Communication with other agents:

- Provides section ownership data to Matching Agent
- Receives section focus data from Interaction Analysis Agent
- Shares assignment updates with Smart Notifications Agent

Prompts:

- "Which team member has the relevant background to handle this section's content?"
 - "Are section assignments balanced or is someone overloaded?"
 - "Should the assigned expert be notified when someone struggles with their section?"
 - "Have all critical sections been assigned to ensure coverage?"
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4. Matching Agent

Description: Finds the best peer helpers for struggling students by matching them with people who understand the same material better.

Goal: Ensure struggling researchers get connected with the most qualified helpers available at that moment.

Features:

- Scores potential helpers from 0-100 based on multiple factors
- Considers helper's understanding score of the relevant section
- Checks helper's current availability and activity status
- Accounts for section assignments (prefer assigned experts)
- Ranks all potential helpers and recommends top 3
- Auto-registers users when they interact with content

Communication with other agents:

- Receives understanding scores from Interaction Analysis Agent
- Sends match recommendations to Discussion Analysis Agent
- Gets section ownership data from Role Assignment Agent
- Provides match results to Smart Notifications Agent

Prompts:

- "What's the helper's demonstrated understanding level for the specific confused section?"
- "Is the potential helper currently active and available, or occupied with their own reading?"

- "Calculate match score: helper understanding (40%) + availability (30%) + section expertise (30%)"
 - "Among available helpers, who has successfully helped with similar confusions before?"
 - "Should we prioritize section-assigned experts over higher-scored general helpers?"
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5. Progress Tracking Agent

Description: Logs the entire learning journey by recording every interaction, struggle, help session, and breakthrough to create a complete history.

Goal: Maintain a comprehensive record of how understanding develops over time for analysis and reflection.

Features:

- Listens to events from all other agents
- Maintains timestamped logs of all user sessions
- Creates journey summaries showing progression
- Records struggle points and how they were resolved
- Tracks help session outcomes and effectiveness
- Generates learning analytics

Communication with other agents:

- Receives events from all other agents (universal listener)
- Provides historical data when requested by any agent

Prompts:

- "What's the complete timeline of this user's struggle: when did it start, what interventions occurred, and when was it resolved?"
 - "How effective was peer help vs AI help for this type of confusion?"
 - "Are there patterns in when users struggle - specific sections, times, or after certain interactions?"
 - "What's the typical time-to-resolution for different types of confusion?"
 - "Which sections generate the most struggles across all users, indicating difficult content?"
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6. AI Tutor Agent

Description: Provides instant explanations using GPT-4 when students highlight confused text and ask for help.

Goal: Offer immediate, personalized explanations when peer help isn't available or when users prefer AI assistance.

Features:

- Receives highlighted confused text from users
- Sends context to OpenAI GPT-4 API with educational prompts
- Returns simplified explanations with concrete examples
- Maintains conversation history for follow-up questions
- Adapts explanation complexity based on user's background
- Provides alternative explanations if first one doesn't help

Communication with other agents:

- Receives confused highlight data from Interaction Analysis Agent

- Works as alternative to Discussion Analysis Agent (peer help)
- Sends tutoring session logs to Progress Tracking Agent

Prompts:

- "What prerequisite knowledge is missing that's causing this confusion?"
 - "Break down this complex concept into 3-4 digestible steps with concrete examples"
 - "Is this a terminology issue, conceptual misunderstanding, or connection to prior knowledge?"
 - "Generate an explanation at the appropriate level: what's the user's apparent background?"
 - "Provide an analogy from everyday experience that maps to this technical concept"
 - "What follow-up question should I ask to verify their understanding?"
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7. Smart Notifications Agent

Description: Sends targeted, context-aware notifications to the right people at the right time using Socket.io.

Goal: Keep users informed about relevant help opportunities without overwhelming them with unnecessary notifications.

Features:

- Generates different notification types based on context
- Delivers "Get AI Help" notifications to struggling users
- Sends "Want to help?" invitations to qualified helpers
- Uses strict user ID targeting to prevent broadcast spam

- Supports bidirectional notifications (struggler + helper)
- Includes relevant context in each notification
- Handles notification delivery via Socket.io

Communication with other agents:

- Receives struggle events from Interaction Analysis Agent
- Gets helper recommendations from Matching Agent
- Triggers chat invitations for Discussion Analysis Agent
- Logs notification delivery to Progress Tracking Agent

Prompts:

- "Has the user been struggling long enough to warrant notification, or should we wait to see if they self-recover?"
 - "Will this notification interrupt the helper's current workflow or is it a good time?"
 - "What contextual information does the helper need to decide if they can help effectively?"
 - "Should this be an immediate notification or queued for when the user finishes their current section?"
 - "Is this user notification-fatigued - have we sent too many in the last 30 minutes?"
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Agent Communication Flow

Complete System Flow: User Struggling → Help (Peer or AI)

1. User struggling with Section 3



2. Interaction Analysis Agent

(detects low understanding: 30/100)



3. [Struggle Event Emitted]



4. Matching Agent

(searches for helpers)

Smart Notifications Agent

(prepares notifications)



Found Peer No Peers

(Score 90) Available



→ 5a. AI Tutor Agent

- Receives confused text

- Sends to GPT-4

- Returns explanation

- User asks follow-ups



Resolution via AI



→ 5b. Discussion Analysis Agent

- Opens peer chat

- Generates conversation starter

- Facilitates discussion



Resolution via Peer



6. Progress Tracking Agent

(logs entire journey: struggle → helper type → resolution)

What's Working Now

- Real-time highlight synchronization across all users
 - Understanding detection triggers correctly
 - Peer matching finds helpers with 90+ match scores
 - Bidirectional notifications deliver to correct users
 - Chat system with invitation flows working
 - AI tutor responds to confused highlights
 - Journey logging captures all events
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Current Module Status

Module #	Module Name	Status
2	Interaction Analysis	✓ Complete
3	Discussion Analysis	90% (chat integration working)
4	Role Assignment	✓ Complete
5	Matching	✓ Complete
1	Reflection Analysis	Not started
6	Fact Checking	Not started
7	Related Work Analysis	Not started
8	Structured Paper Summary	✓ Complete

Module #	Module Name	Status
9	Progress Tracking	✓ Complete

Next Steps (Next 2 Weeks)

Week 1:

- Complete Module 3 (Discussion Analysis) - finish Socket.io integration
- Start Module 1 (Reflection Analysis) - paragraph highlighting system
- Begin Module 6 (Fact Checking) - API integration

Week 2:

- Continue Module 1 (Reflection Analysis) and Module 6 (Fact Checking)
- Start Module 7 (Related Work Analysis) - paper similarity detection